

Unexpectedness by Design: Causal Serendipity Optimization for Implicit Recommender Systems

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Overview



- Motivating Question
- Existing Literature & Methodology
- Applied Algorithm
- Discussion

Motivating Question

Problem Statement

*How do we build an **implicit recommender system** that incorporates the concept of **serendipity**?*

Recall: Matrix Factorization

- MF represents users as rows and items as columns; ratings in cells
- Traditional MF tries to discover hidden (latent) features about both the users and items:

Definition (Matrix Factorization)

1. Breaks sparse matrix into two smaller, dense matrices: (1) a **user matrix** and (2) an **item matrix**
2. To predict what user A would rate item B , MF takes the **dot product** of user A 's and item B 's latent vectors. If their vectors align, the predicted rating is high
3. MF determines the values of the latent vectors by comparing prediction for observed to actual rating, measuring the error, and minimizing with respect to a loss function

Recall: Matrix Factorization

Step 3 aims to minimize MSE:

$$\text{Minimize} \quad \sum_{(u,i) \in \text{Observed}} (y_{ui} - \hat{y}_{ui})^2$$

More formally, let the matrix of observed ratings be Y . We represent each user u with a latent preference vector θ_u and each item i with a latent attribute vector β_i . The model predicts a rating via their dot product: $\hat{y}_{ui} = (\theta_u^\top \beta_i)$. Incorporating L2 regularization (Ridge penalty) to prevent overfitting by penalizing excessively large weights, we aim to minimize the objective function:

$$\min_{\Theta, B} \sum_{(u,i) \in \text{Observed}} (y_{ui} - \theta_u^\top \beta_i)^2 + \lambda(||\theta_u||^2 + ||\beta_i||^2)$$

RecSys as a Causal Inference Problem

The Problem: To satisfy the missing completely at random (MCAR) property, it must hold that treatment assignment, or exposure, is independent of the potential outcomes. That is:

$$a_{ui} \perp Y_{ui}(1), Y_{ui}(0)$$

- Wang et al. argue that this violation of MCAR serves as a form of confounding bias: the variables that cause the data to be missing (e.g., a user's genre preferences, location, affinity for a director) serve as unobserved confounders that influence the final rating.
- **Importantly:** Matrix factorization requires MCAR to hold!

RecSys as a Causal Inference Problem

The Solution: Fit Poisson factorization to the binary exposure matrix a_{ui} to determine probabilities of a user naturally encountering a movie. Incorporate outputted expected probability exposure $\hat{a}_{ui}$ into the final outcome model to serve as a substitute confounder.

- Rather than predicting the rating via a simple dot product (the traditional MF prediction, $\theta_u^\top \beta_i$), Wang et al. modify the outcome equation to condition on this exposure probability:

$$y_{ui}(a) = \theta_u^\top \beta_i \cdot a + \gamma_u \cdot \hat{a}_{ui} + \epsilon_{ui}$$

- γ_u is a user-specific coefficient that absorbs the bias of the exposure mechanism
- By absorbing confounding bias into the $\gamma_u \cdot \hat{a}_{ui}$ term, the remaining latent vectors ($\theta_u^\top \beta_i$) isolate the true, unconfounded user preference

Modeling Implicit Feedback



- Wang et al.'s causal inference framework provides a strong theoretical foundation for serendipity in the context of a recommender system
- **Problem:** It operates the assumption of known, explicit outcomes (e.g., 5-star ratings)
- **Solution:** Johnson's framework for **logistic matrix factorization** proposes that raw interaction volume (e.g., stream counts), denoted as r_{ui} , can be treated as a measure of confidence, which can stand in for an explicit rating. From the implicit data, Johnson defines a confidence metric:

$$c_{ui} = \alpha r_{ui}$$

where α is a scaling hyperparameter. As r_{ui} increases, the system's confidence that the user genuinely prefers the item increases

Logistic Matrix Factorization

- **Problem:** Rather than minimizing MSE, use logistic loss optimization to treat unobserved interactions as low-confidence negatives, rather than zeroes:

$$p(l_{ui}) = \sigma(\theta_u^\top \beta_i)$$

where $p(l_{ui})$ is the probability that a user likes an item and σ is the standard logistic sigmoid function

- To combine this with our causal objective, we inject Wang's substitute confounder directly into the logistic equation, yielding our causally-conditioned logistic prediction:

$$p(l_{ui}) = \sigma(\theta_u^\top \beta_i + \gamma_u \hat{a}_{ui})$$

Defining Serendipity

- The concept of serendipity is not well defined, but modern definitions are succinctly summarized by Kotkov
- We propose that a serendipitous recommendation is one that is both surprising and satisfying
- It is particularly unclear how to measure or predict this under a traditional RecSys framework, but our causal framework allows us to frame it as the value of an intervention

Definition (Serendipity)

We determine a recommendation to be **serendipitous** if it is an item that a user has a high true preference for (and will thus rate highly), *but for which the user has a very low probability of discovering without algorithmic intervention.*

Defining Serendipity

We can map these two requirements directly to the outputs of our causal model:

1. **Relevance (Likely to Like):** Represented by the unconfounded baseline preference, $\theta_u^\top \beta_i$. Because the exposure bias was absorbed by γ_u , this dot product represents the user's pure affinity for the item.
2. **Expectedness (Likely to Encounter):** Represented by the substitute confounder, $\hat{a}_{ui}$. This captures the probability that the user would naturally discover the item through mainstream popularity or existing habits (the confounding variables we provided examples for earlier).

Defining Serendipity

From these requirements, we construct and propose a **Serendipity Score** (S_{ui}) for unseen items by maximizing unconfounded relevance while penalizing expectedness:

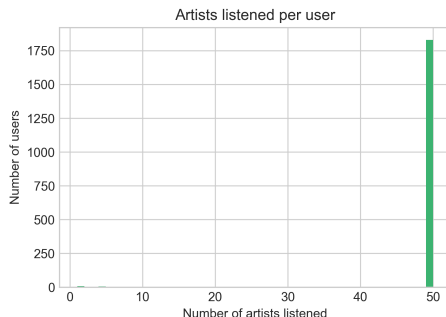
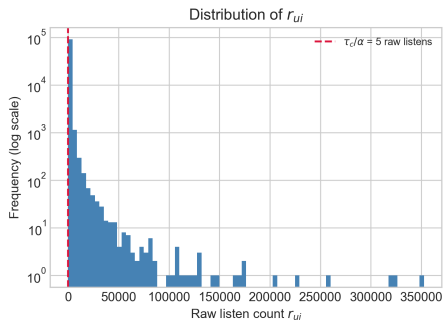
$$S_{ui} = \frac{\theta_u^\top \beta_i}{\hat{a}_{ui} + \epsilon}$$

where ϵ is a small constant to prevent division by zero for items with near-zero natural exposure.

Last.fm HetRec 2011 Dataset

This dataset contains social networking, tagging, and music artist listening information.

- 1,892 Users, 17,632 Artists, 92,834 Total Interactions
- Median Listens: 260
- Maximum Listens: 352,698 (*Extreme Right Skew*)



Case Study: User 5 Profile

User 5's Top 10 Most-Listened Artists

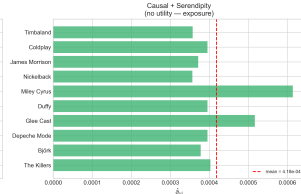
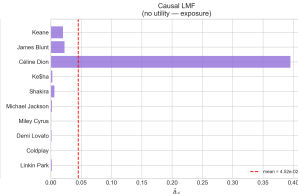
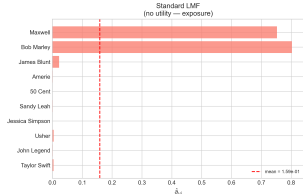
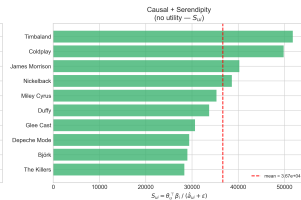
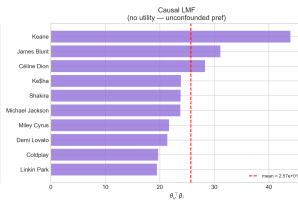
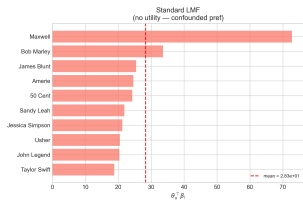
Rank	Artist Name	Total Listens
1	Rihanna	43,864
2	Britney Spears	11,112
3	Jordin Sparks	6,266
4	Lady Gaga	5,846
5	Kelly Clarkson	5,379
6	Mariah Carey	5,188
7	Christina Aguilera	4,201
8	Ashlee Simpson	4,059
9	Leona Lewis	3,251
10	Kylie Minogue	3,151

Model Comparison for User 5

- **Standard LMF:** What is popular?
- **Causal LMF:** What does the user genuinely prefer?
- **Causal + Serendipity:** What would surprise the user?

Model Comparison for User 5 Cont.

Top 10 Recommended Artists for User 5 (for Each Model)



Limitations and Further Applications

Training Limitations

- The dataset caps each user to 50 artists, restricting the scope the models can train on.
- Implicit data is ambiguous.

Future Scope

- Additional hyperparameter tuning to better fit this dataset.
- Testing the model across a much wider range of users to make sure it generalizes beyond a single case.

Thank You!

References

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Appendix: RecSys as a Causal Inference Problem

Definition (Potential Outcomes)

In terms of the Neyman-Rubin causal model, every user-item pair has two **potential outcomes**:

- $Y_{ui}(1)$: The rating user u would give item i if they were exposed to it (e.g., forced to watch a movie)
- $Y_{ui}(0)$: The rating the user would give the item i if they were not exposed to it
- $a_{ui} \in \{0, 1\}$: The treatment assignment (e.g., whether they actually watched the movie)

Note: We only observe the factual outcome: $Y_{ui} = Y_{ui}(a_{ui})$.

Appendix: The Utility of Utility Constraints

Problem: Simply ranking recommendations by S_{ui} in a descending order risks presenting a slate of obscure content to the user.

Solution: Peng et al. propose a consumption constraint framework:

Definition (Utility (Peng et al.))

Define **utility**, denoted as $util_1$ as the probability that the user likes at least one item in the recommended set. Grouping candidate items into m categories, where category t has conditional user satisfaction rate q_t , and n_t represents the number of slots allocated to that category, the utility of the slate is defined as the complement of the user disliking everything:

$$util_1 = 1 - \prod_{t=1}^m (1 - q_t)^{n_t}$$

To maximize $util_1$, the recommender system must allocate its N recommendation slots inversely proportional to a category's satisfaction